<u>Assignment-4</u>

Take one domain and build business Understanding

Business Understanding Phase – Data Science Project

Domain Chosen: Online Retail (E-Commerce Platform)

1. Introduction

In any data science or analytics project, the first and most critical phase is **Business Understanding**. Before collecting data, building models, or performing analysis, it is essential to clearly understand the business problem, objectives, constraints, and expected outcomes. Without this stage, even technically advanced models may fail to deliver value because they do not address real business needs.

Business understanding ensures that the project aligns with organizational goals, solves meaningful problems, and produces actionable insights. It helps define the purpose of analysis, identify stakeholders, clarify success criteria, and determine how results will be used. In this assignment, we explore the business understanding phase using a real-world domain example: an **online retail (e-commerce) company**.

2. Domain Overview – Online Retail Industry

The online retail industry is one of the fastest-growing sectors worldwide. E-commerce platforms sell products through websites or mobile applications, allowing customers to browse, compare, and purchase items online. Examples of services provided by online retail platforms include:

- Product search and browsing
- Personalized recommendations
- Online payments
- Order tracking
- Customer reviews

Because these platforms operate digitally, they generate massive amounts of data such as customer behavior, purchase history, product views, and browsing patterns. This makes the domain highly suitable for data analysis and machine learning applications.

3. What is Business Understanding?

Business understanding is the process of defining the problem from a business perspective rather than a technical perspective. It answers questions such as:

- What problem is the company trying to solve?

- Why is this problem important?
- Who are the stakeholders?
- What results are expected?
- How will success be measured?

It ensures that the project focuses on solving the right problem instead of just analyzing data without purpose.

4. Business Problem Definition

Suppose an online retail company is facing the following issue:

Customers visit the website but do not complete purchases.

This situation is known as **low conversion rate**. A conversion rate represents the percentage of visitors who make a purchase.

Problem Statement

The company wants to increase its conversion rate by understanding customer behavior and predicting which visitors are most likely to purchase products.

5. Business Objectives

The primary objective of the project is to increase sales revenue by improving customer conversion.

Specific Goals

- Identify factors influencing purchase decisions
- Predict which customers are likely to buy
- Recommend products to users
- Reduce cart abandonment
- Improve customer experience

6. Stakeholders Involved

A stakeholder is anyone affected by or involved in the project.

Key Stakeholders

- Business managers → want higher revenue
- Marketing team → wants better targeting
- Data scientists → build predictive models

- Website designers → improve user experience
- Customers → expect relevant recommendations

Understanding stakeholder expectations is crucial because each group has different goals.

7. Business Constraints

Every project operates under limitations. These constraints must be identified early.

Common Constraints

- Limited budget
- Time restrictions
- Data availability
- Privacy regulations
- Computing resources
- Technical expertise

For example, if customer data cannot be shared due to privacy laws, analysis methods must comply with legal requirements.

8. Business Success Criteria

Success criteria define how the organization measures project success.

Examples:

- Increase conversion rate by 15%
- Increase average order value
- Reduce cart abandonment by 20%
- Improve customer retention

Clear success criteria help evaluate whether the project achieved its goals.

9. Data Science Goals

After defining business goals, they must be translated into data science goals.

Business Goal: Increase sales
Data Goal: Predict customer purchase probability

This conversion ensures that technical work directly supports business objectives.

10. Analytical Approach

To solve the business problem, analysts may choose different methods such as:

- Classification models
- Recommendation systems
- Customer segmentation
- Predictive analytics

For example, a machine learning classification model can predict whether a visitor will buy a product or not.

11. Required Data

To perform analysis, the following types of data may be needed:

- Customer demographics
- Browsing history
- Purchase history
- Time spent on pages
- Device type
- Location
- Cart activity

The quality and availability of data strongly influence project success.

12. Risks and Challenges

Every business project involves potential risks.

Possible risks include:

- Incomplete or missing data
- Biased datasets
- Privacy concerns
- Model inaccuracies
- Changing business requirements

Identifying these risks early helps plan solutions and avoid project failure.

13. Business Impact

If the project succeeds, it can produce significant business benefits:

- Increased revenue
- Better customer experience
- Improved marketing efficiency
- Reduced operational costs
- Stronger customer loyalty

These impacts justify investment in data science projects.

14. Example Scenario

Imagine a user visits an online store and views several products but leaves without buying anything. A predictive model can analyze this behavior and determine whether the user is likely to return and purchase. If the probability is high, the system may send a discount coupon or personalized recommendation. This targeted approach increases the chances of conversion.

15. Importance of Business Understanding Phase

The business understanding phase is crucial because it:

- Defines clear objectives
- Aligns technical work with business goals
- Prevents wasted effort
- Reduces project risk
- Improves decision-making
- Ensures useful results

Many data science projects fail not because of technical issues, but because the business problem was not properly understood.

16. Relationship with Data Science Lifecycle

Business understanding is the first stage of the data science lifecycle, which usually includes:

1. Business understanding
2. Data collection
3. Data preparation
4. Modeling
5. Evaluation
6. Deployment

If the first stage is weak, all later stages become ineffective.

17. Advantages of Strong Business Understanding

- Clear direction for analysis
- Efficient use of resources
- Better stakeholder communication
- Higher success probability
- Meaningful insights

18. Limitations if Ignored

If business understanding is skipped:

- Wrong problem may be solved
- Results may be useless
- Time and money wasted
- Stakeholders dissatisfied
- Project may fail

Conclusion

The business understanding phase is the most important step in any data science or analytics project because it establishes the foundation upon which all other stages depend. By clearly defining the business problem, identifying stakeholders, setting objectives, and determining success criteria, organizations ensure that their analytical efforts are meaningful and aligned with real-world goals. In the example of an online retail company, understanding why customers do not complete purchases allows analysts to design targeted solutions such as predictive models and personalized recommendations.

Without proper business understanding, even sophisticated algorithms and advanced analytics may fail to deliver value. Therefore, successful data-driven organizations prioritize this phase before beginning technical work. A strong business understanding not only improves decision-making but also maximizes the impact of data science projects, leading to better performance, increased revenue, and enhanced customer satisfaction.